

Exact closedness of a width-two sparse matrix-product image

Candidate partial result for arXiv:2306.02666

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Status. `candidate_partial_likely_valid`. This packet proves an exact hidden-width-two sub-case of the efficient closedness problem and gives an exact Bernoulli probability for the resulting nonclosedness certificate. It does not solve the unrestricted-width problem.

Abstract

For two coordinate rectangles $S_i \times T_i$, consider the set of sums of one rank-one matrix on each rectangle. We give a necessary and sufficient combinatorial criterion for this set to be closed. Besides the familiar local LU degeneration, exactly two transposed “coalescing direction” degenerations occur. The criterion is linear-time after the supports are read. As an upgrade, we derive the exact probability of nonclosedness for independent Bernoulli supports. Through Theorem 3.1 of the source paper, every nonclosed pattern certifies a sparse ReLU architecture for which some finite regression problem has no optimum.

1 Source question

The source is Q.-T. Le, E. Riccietti, and R. Gribonval, *Does a sparse ReLU network training problem always admit an optimum?*, arXiv:2306.02666v2 [1]. Section 3.3 asks for more efficient algorithms deciding the closedness of the sparse matrix-product image $\mathcal{L}_{\mathbf{I}}$ and leaves the general problem open. Question 3.2 asks:

If the supports of the weight matrices are randomly sampled from a distribution, what is the probability that the corresponding training problem potentially admits no optimal solutions?

The source then gives a polynomial sufficient detector for one-hidden-layer networks, explicitly noting that it may have false negatives. Figure 1 reproduces the full source context from page 7.

2 Setup

For $S \subseteq [m]$ and $T \subseteq [n]$, define the coordinate rank-one cone

$$\mathcal{C}(S, T) = \{uv^\top : u \in \mathbb{R}^m, v \in \mathbb{R}^n, \text{supp}(u) \subseteq S, \text{supp}(v) \subseteq T\}.$$

For two hidden components, the source’s two-factor linear image is precisely

$$\mathcal{L} = \mathcal{C}(S_1, T_1) + \mathcal{C}(S_2, T_2). \tag{1}$$

Put

$$S_0 = S_1 \cap S_2, \quad T_0 = T_1 \cap T_2, \quad S_i^\circ = S_i \setminus S_{3-i}, \quad T_i^\circ = T_i \setminus T_{3-i}.$$

3.3 The closedness of $\mathcal{L}_I$ is algorithmically decidable

Theorem 3.1 leads to a natural question: given I , how to check the closedness of $\mathcal{L}_I$, a subset of $\mathbb{R}^{N_L \times N_0}$. To the best of our knowledge, there is not any study on the closedness of $\mathcal{L}_I$ in the literature. It is, thus, not known whether deciding on the closedness of $\mathcal{L}_I$ for a given I is polynomially tractable. In this work, we show it is at least decidable with a doubly-exponential algorithm. This algorithm is an application of *quantifier elimination*, an algorithm from real algebraic geometry [2].

Lemma 3.3. *Given $I = (I_L, \dots, I_1)$, the closedness of $\mathcal{L}_I$ is decidable with an algorithm of complexity $O((4L)^{C^{k-1}})$ where $k = N_L N_0 + 1 + 2 \sum_{i=1}^L |L_i|$ and C a universal constant.*

We prove Lemma 3.3 in Appendix B.4. Since the knowledge of I is usually available (either fixed before training [19, 31, 4, 21, 3] or discovered by a procedure before re-training [10, 14, 32]), the algorithm in Lemma 3.3 is able to verify whether the training problem might not admit an optimum. While such a doubly exponential algorithm in Lemma 3.3 is seemingly impractical in practice, small toy examples (for example, Example 3.1 with $d = 2$) can be verified using Z3Prover, a software implementing exactly the algorithm in Lemma 3.3. However, Z3Prover is already unable to terminate when run on the LU architecture of Example 3.1 with $d = 3$. This calls for more efficient algorithms to determine the closedness of $\mathcal{L}_I$ given I . The same algorithmic question can be also asked for $\mathcal{F}_I$. We leave these problems (in this general form) as open questions.

In fact, if such a polynomial algorithm (to decide the closedness of $\mathcal{L}_I$) exists, it can be used to answer the following interesting question:

Question 3.2. *If the supports of the weight matrices are randomly sampled from a distribution, what is the probability that the corresponding training problem potentially admits no optimal solutions?*

While simple, this setting does happen in practice since random supports/binary masks are considered a strong and common baseline for sparse DNNs training [22]. Thanks to Theorem 3.1 if $\mathcal{L}_I$ is not closed then the support is “bad”. Thus, to have an estimation of a *lower bound* on the probability of “bad” supports, we could sample the supports from the given distribution and use the polynomial algorithm in question to *decide* if $\mathcal{L}_I$ is closed. Unfortunately, the algorithm in Lemma 3.3 has doubly exponential complexity, thus hindering its practical use. However, for one-hidden-layer NNs, there is a *polynomial* algorithm to *detect* non-closedness: intuitively, if the support constraint is “locally similar” to the LU structure, then $\mathcal{L}_I$ is not closed. This result is elaborated in Appendix B.5 and Lemma B.8. The resulting detection algorithm can have false negatives (i.e., it can fail to detect more complex configurations where $\mathcal{L}_I$ is not closed) but no false positive.

Figure 1: Source page 7: the efficient-decision problem, Question 3.2, and the false-negative limitation of the existing detector.

3 Proof intuition

Any escaping factor sequence must cancel its leading rank-one term inside the common rectangle $S_0 \times T_0$. There are only three ways for the finite first-order residue to become impossible to express by two limiting rank-one terms. One rectangle can carry an exclusive row and column, producing the usual 2×2 LU anti-diagonal. Alternatively, the two rank-one column directions can coalesce while two exclusive columns pin both of them to the same limiting line, leaving a transverse common column; the transpose gives the third case.

If none of those mechanisms is available, a short set-theoretic case split forces the rectangles to be disjoint, cross-nested, or one-dimensional in the only noncomparable direction. The cross-nested image is exactly a determinantal set described by three rank inequalities, so it is closed. This also explains why the classification yields an immediate decision algorithm. The original general-width problem remains open because three or more cones can fill one another’s boundary points or support genuinely multi-term cancellations.

4 Exact classification

Theorem 1 (Hidden width two). *The set $\mathcal{L}$ in (1) is nonclosed if and only if S_0 and T_0 are nonempty and at least one of the following holds:*

- (i) S_1°, T_1° are nonempty, or S_2°, T_2° are nonempty;
- (ii) $|S_0| \geq 2$ and both T_1°, T_2° are nonempty;
- (iii) $|T_0| \geq 2$ and both S_1°, S_2° are nonempty.

Consequently, closedness can be decided in $O(m+n)$ time from bit-vector representations of the four supports.

We first record the elementary closed-case lemma used in the converse.

Lemma 1 (Cross-nested block lemma). *Suppose $S_2 \subseteq S_1$ and $T_1 \subseteq T_2$. Put $A = S_1 \setminus S_2$, $C = S_2$, $B = T_2 \setminus T_1$, and $D = T_1$. After ordering rows as A, C and columns as B, D , the image $\mathcal{L}$ is exactly the set of block matrices*

$$M = \begin{pmatrix} 0 & Y \\ X & Z \end{pmatrix} \quad \text{such that} \quad \text{rank } M \leq 2, \quad \text{rank } X \leq 1, \quad \text{rank } Y \leq 1. \quad (2)$$

In particular, $\mathcal{L}$ is closed.

Proof. Every element of $\mathcal{L}$ is the sum of a rank-one matrix supported on $(A \cup C) \times D$ and one supported on $C \times (B \cup D)$, so it satisfies (2).

Conversely, first suppose X and Y are nonzero. The columns over B span a line V contained in the C coordinate subspace. Because $Y \neq 0$, some column over D is not in V . Since $\text{rank } M \leq 2$, this column together with V spans the entire column space. Hence every D -column is the sum of a multiple of one fixed vector in $A \cup C$ and a multiple of the generator of V , while every B -column uses only the latter. These are the required two rank-one summands.

If $X \neq 0$ and $Y = 0$, the same argument applies with all column vectors in the C coordinate subspace: the B -columns span one line and the total column space has dimension at most two. The case $X = 0$, $Y \neq 0$ follows by applying this argument to the transpose. If $X = Y = 0$, then Z has rank at most two and is the sum of two rank-one matrices supported on the common rectangle $C \times D$. Thus every matrix in (2) belongs to $\mathcal{L}$. The right-hand side of (2) is cut out by support equations and minors, hence is closed. $\square$

Proof of Theorem 1. Assume first that (i) holds for the first rectangle. Choose $a \in S_1^\circ$, $c \in S_0$, $b \in T_1^\circ$, and $d \in T_0$. For $t > 0$,

$$(te_c + e_a)(e_d + t^{-1}e_b)^\top + (-te_c)e_d^\top \longrightarrow e_ce_b^\top + e_ae_d^\top =: M. \quad (3)$$

The two summands on the left belong to the two cones. On rows a, c and columns b, d , the second cone can contribute only at (c, d) , while a rank-one first summand cannot match the anti-diagonal entries of M . Thus $M \notin \mathcal{L}$. The other half of (i) is symmetric.

For (ii), choose distinct $p, q \in S_0$, columns $b \in T_1^\circ$, $e \in T_2^\circ$, and $d \in T_0$. Then

$$e_p(e_b - te_d)^\top + (e_p + t^{-1}e_q)(e_e + te_d)^\top \longrightarrow e_pe_b^\top + e_pe_e^\top + e_qe_d^\top =: N. \quad (4)$$

If N were in $\mathcal{L}$, its exclusive b -column would force the first rank-one column direction to be $\mathbb{R}e_p$, and its exclusive e -column would force the second direction to be $\mathbb{R}e_p$. Every common column

would then lie in $\mathbb{R}e_p$, contradicting the d -column e_q . Hence $N \notin \mathcal{L}$. Case (iii) is the transpose of this construction. These sequences prove nonclosedness in all listed cases.

Conversely, suppose S_0, T_0 are nonempty and none of (i)–(iii) holds. If both S_1° and S_2° are nonempty, failure of (i) forces $T_1 = T_2$; failure of (iii) then forces $|T_0| = 1$. In that case both cones lie in a one-dimensional column coordinate space, and their sum is a linear subspace. The transposed argument treats the case in which both T_1°, T_2° are nonempty.

In every remaining case, after possibly interchanging indices, the supports are cross-nested: $S_2 \subseteq S_1$ and $T_1 \subseteq T_2$. Lemma 1 shows that their sum is closed. If S_0 or T_0 is empty, the two cones occupy disjoint coordinate sets, so coordinate projection separates two convergent summands and their sum is closed. This exhausts all possibilities and proves the converse. $\square$

Example 1 (A minimal false negative of the local-LU detector). *Let $S_1 = S_2 = \{1, 2\}$, $T_1 = \{1, 3\}$, and $T_2 = \{2, 3\}$. The source detector cannot find three entries exclusive to one rank-one rectangle around a shared corner, because the row supports coincide. Nevertheless (ii) applies, and*

$$e_1(e_1 - te_3)^\top + (e_1 + t^{-1}e_2)(e_2 + te_3)^\top \longrightarrow e_1e_1^\top + e_1e_2^\top + e_2e_3^\top$$

is a missing boundary point. Thus the sufficient detector's admitted false negatives already occur with two hidden components.

5 Upgrade: exact Bernoulli probability

Let every row membership in S_1, S_2 be independent Bernoulli(α), and every column membership in T_1, T_2 independent Bernoulli(β), with the row and column masks independent. For a row, let $\rho = (\rho_{11}, \rho_{10}, \rho_{01}, \rho_{00})$ count the four membership types across the m rows. Define τ analogously across the n columns. For $p \in [0, 1]$ and $\kappa_\bullet$ summing to k , put

$$w_{k,p}(\kappa) = \frac{k!}{\prod_z \kappa_z!} (p^2)^{\kappa_{11}} [p(1-p)]^{\kappa_{10}+\kappa_{01}} [(1-p)^2]^{\kappa_{00}}.$$

Let $\Phi(\rho, \tau)$ be the Boolean condition

$$\begin{aligned} \rho_{11} \geq 1, \quad \tau_{11} \geq 1, \quad \text{and} \\ [(\rho_{10} \geq 1 \wedge \tau_{10} \geq 1) \vee (\rho_{01} \geq 1 \wedge \tau_{01} \geq 1) \vee (\rho_{11} \geq 2 \wedge \tau_{10} \geq 1 \wedge \tau_{01} \geq 1) \\ \vee (\tau_{11} \geq 2 \wedge \rho_{10} \geq 1 \wedge \rho_{01} \geq 1)]. \end{aligned} \tag{5}$$

Corollary 1. *Under this model, the exact probability that $\mathcal{L}$ is nonclosed is*

$$\boxed{\sum_{\rho_{11}+\rho_{10}+\rho_{01}+\rho_{00}=m} \sum_{\tau_{11}+\tau_{10}+\tau_{01}+\tau_{00}=n} w_{m,\alpha}(\rho) w_{n,\beta}(\tau) \mathbf{1}_{\Phi(\rho,\tau)}}. \tag{6}$$

It can be evaluated in $O(m+n)$ arithmetic operations by retaining only the threshold states $0, 1, \geq 2$ for common counts and $0, \geq 1$ for exclusive counts. Every mask counted in (6) is potentially ill-posed for a suitable finite regression problem in the sense of the source paper.

Proof. The four row categories have probabilities $\alpha^2, \alpha(1-\alpha), \alpha(1-\alpha), (1-\alpha)^2$, so ρ has the displayed multinomial mass; the column count has the analogous independent law with parameter β . Condition (5) is exactly Theorem 1 written in terms of these counts, proving the sum. The finite-state dynamic program follows because no larger count is distinguished by Φ . Finally, Theorem 3.1 and Proposition 3.1 of the source turn nonclosedness of $\mathcal{L}$ into nonclosedness of a finite realization set and hence a regression instance whose infimum is not attained. $\square$

6 Verification, novelty, and limitations

The accompanying script exhaustively checks that the theorem’s nonclosed and proved-closed cases partition all support patterns for $m, n \leq 4$. It also checks formula (6) against direct weighted enumeration for $m, n \leq 3$ and 27 dimension/parameter choices. These computations verify the case split and bookkeeping, not the proof.

A bounded primary-source search on 11 August 2026 used the exact source title, the phrases “rank-one supports closedness” and “sum of two rank-one cones”, and the source’s cited fixed-support factorization paper arXiv:2112.00386 [2]. It found the source’s sufficient local-LU criterion but no exact hidden-width-two converse or the two coalescing-direction degenerations. Novelty confidence is moderate because this elementary result may exist under other terminology.

The main limitation is structural: for hidden width at least three, a third cone can fill a two-cone boundary point or join a multi-term cancellation. Also, $\mathcal{L}$ closed does not imply that every ReLU realization set is closed. Corollary 1 therefore gives the exact probability of the source’s linear nonclosedness certificate at width two, but only a lower bound for the probability of all potentially ill-posed ReLU masks.

Human-review recommendation. Recommended for expert review as a fully proved substantial partial result. The key checks are Lemma 1, exhaustion of the support case split, and possible prior appearance of the classification in matrix-factorization geometry.

References

- [1] Q.-T. Le, E. Riccietti, and R. Gribonval, *Does a sparse ReLU network training problem always admit an optimum?*, Advances in Neural Information Processing Systems 36 (2023), arXiv:2306.02666.
- [2] Q.-T. Le, E. Riccietti, and R. Gribonval, *Spurious Valleys, NP-hardness, and Tractability of Sparse Matrix Factorization With Fixed Support*, SIAM J. Matrix Anal. Appl. 44 (2023), 503–529, arXiv:2112.00386.